<u>Outcome measurement in Health Care:</u> QALY measurement

In CUA, health outcomes are measured in Quality Adjusted Life Years (QALYs) gained. The results are expressed as a cost per QALY gained. By converting the effectiveness data to a common unit of measurement, QALYs gained, CUA is able to incorporate both the increase in the Q_oL (reduced morbidity) and the increase in the quality of life (reduced mortality), which are the overriding objects most health care programs stipulate.

Dr. Fryback (a physician & health economist) once remarked in 1993 that…“the healthcare industry is a $900+ billion endeavor that does not know how to measure its mean product: Health”. The quotation seems correct and reveals the undignified position of outcome research in health economics.

If we cannot measure “health outcome” or “health benefits” from a particular health program based on individual preferences or utilities, how do we purify allocative efficiency of scarce resources among all competing alternatives? Setting priorities in health care industry such that we can maximize “value for money” essentially means that we can achieve more desirable health benefits from the given budget.

<u>Preference or Utility Assessment:</u> Why it is needed?

Programs in health care have varying objectives:

- The objective of prenatal care might be a reduction in infant mortality;
- Rheumatologists strive to make their patients more functional;
- Screening for breast cancer (mammography) or for prostate cancer leads to early detection and treatment;
- Primary care provides after focus on shortening the cycle of acute illness.

All of these providers are attempting to improve the health of these patients. However, each of them measures health in a different way. Comparing the productivity of a rheumatologist with that of a neonatologist may be like comparing apples and oranges.

The diversity of outcomes to health care had led many analysts to focus on the simplest common ground, typically mortality or life expectancy (LE). Those who are alive are statistically coded as 1 and those who are dead are statistically coded as 0.

Mortality allows the comparison between different diseases. The difficulty is that everyone who remains alive is given the same score. A person confined to bed with irreversible coma is alive and is counted the same as someone who is actively playing volleyball in a picnic. Utility assessment allows the quantification of levels of wellness on the continuum anchored by death and optimum function.

Conceptual Framework

To evaluate HRQ$_0$L, one must consider all the different ways that illness and its treatment affect outcomes. Health concerns can be reduced to two categories: survival duration and Q$_0$L. And determining how illness or treatment affects the desirability of life is a matter of preference, rational choice or utility.
Nearly all HRQ$_0$L measures have multiple dimensions, such as pain and lack of mobility. The exact dimensions vary from measure to measure. There is considerable debate in the field about which dimensions should be included: For example, the most commonly included dimensions are physical functioning, role functioning and mental health.

Relative importance of dimensions

Most treatments have side effects as well as benefits. Generally, the frequencies of various side effects are tabulated. Thus a medication to central high BP might be associated with low probabilities of dizziness, tiredness, impotence and shortness of breath. Should the patient who feels sleepy discontinue the medication? Should a patient with Insulin Dependent Diabetes Mellitus (IDDM) discontinue therapy because he/she develops skin problems at the injection sites? Clearly, local irritation is a side effect of treatment. But without treatment the patient would die. How we should place these side effects within the perspective of total health? Ultimately, we must decide whether treatment produces a net benefit or a net deficit in health status.

The QALY concept

The concept of QALY was first introduced in 1968 by Herbert Klarman and his colleagues in a study on Chronic Renal Failure. The noted that the Q$_0$L with kidney transplant was better than with dialysis, and they estimated a 20% improvement. The cost per life-year gained by different treatment options was also calculated with and without this Q$_0$L adjustment for showing the estimated 20% difference(improvement). Although they did not use the term QALY, but essentially the concept was identical. The term "QLAY" was first popularized by a landmark paper from Harvard, published in NEJM (Weinstein & Stason,1977).

The advantage of QALY as a measure of health outcome is that it can simultaneously capture gains from reduced morbidity and reduced mortality, and combine them into a one-dimensional measure such that more is better. The combination is based on relative desirability of different outcomes.

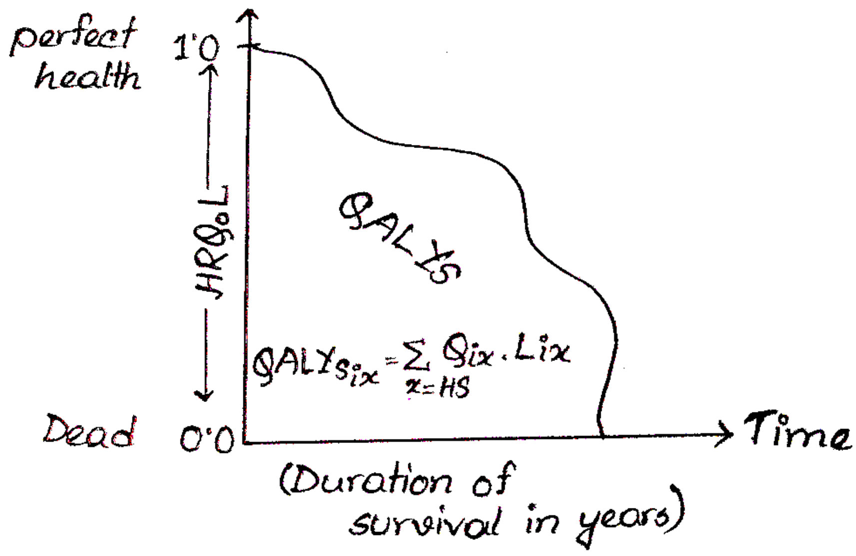


Figure-01: QALY measurement

Figure -1 above depicts the QALY as a measure of health benefits. The continuum is labeled at the top by an optimal/perfect level of HRQ_0L=1.0 and at bottom by a level of HRQ_0L equivalent to dead=0.0. The area under the curve measures the QALYs. A number of terms have been coined to describe this area such as Well Years of Healthy life, QALE, Health Adjusted Person Years, HALE etc.

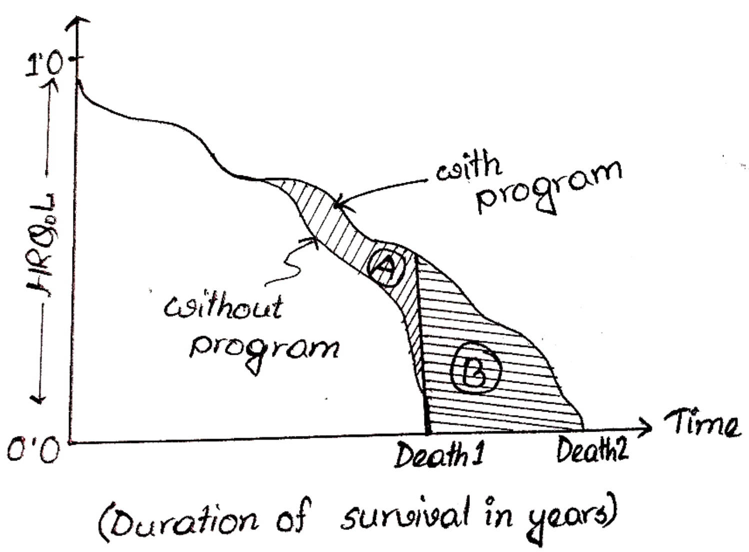


Figure 02: QALYs gained after intervention

The area between the two curves is the number of QALYs gained by this intervention. The area A is the amount of QALY gained due to quality improvement and area B is the amount of QALY gained due to the quality improvement weighted by the HRQ_0L.

Theoretical Foundation of QALY Measures

The main disagreement is not over the QALY concept but rather over how the weights of cases between 0 to 1 are obtained. The calculations of QALY weights are based on the following notions:

a) based on rational preference structure
b) anchored on the concept of perfect/optimal health and dead, and
c) measured on an interval scale (0 to 1)

a) It is based on preferences because this way more desirable health status receive greater weight and will be favoured in the analysis.
b) The QALY weights must contain two reference points on the scale where one measures perfect/optimal health and the other measures death. The conditions worse than death may have negative values.
c) The scale of HRQ_0L (0-1) is an interval scale not a ratio scale. A gain from 0.2 to 0.4 on the scale represents the same increase in desirability as a gain from 0.6 to 0.8. This is required because in QALY calculation these two types of gains appear equal. Interval scale is a cardinal scale, where all parametric statistical calculations are allowed (e.g., mean, sd, t-test, ANOVA etc).

QALY Calculations

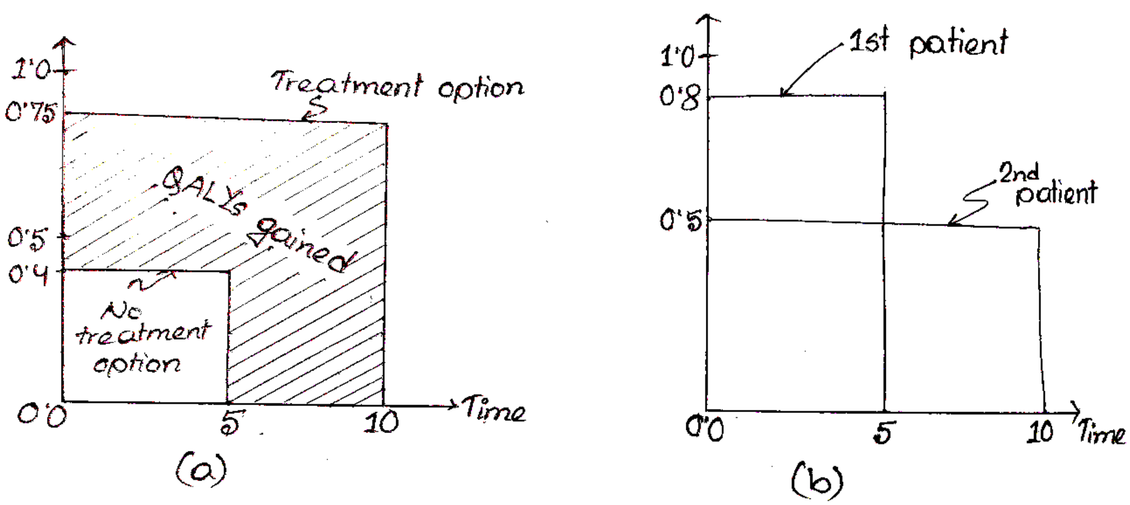


Figure 3: QALY calculations

In figure 3(a), we have QALY measure for the following options:

1. Treatment Option: 10 years*0.75=7.5 QALYs
2. No treatment Options: 5 years*0.40=2.0 QALYs

QALYs gained=5.5 QALYs

If we use 5% discount rate, PV of 5.5 QALYs gained is [$5.5*\frac{1}{(1+.05)^5}$] = 4.31

We could also describe how a patient's Q_0L changes over time. Figure 3(b) shows such profiles for two hypothetical patients.

1st patient: 5 years*0.8=4.0 QALYs

2nd patient: 10 years*0.5=5.0 QALYs

Although the second patient-person survives twice as long as the first, the difference in their QALY is much less.

How are the Utility Values determined?

There are three distinct approaches in obtaining utility values or weights for health status in CUA:

1. Use judgment to estimate the utility values, or a range of values, and then to undertake sensitivity analysis to explore the implications of the judgment. Judgments can be simple estimates made by the analyst himself or by a few physicians or a panel comprising health experts.
2. Use existing utility values available in the literature. Of course, it would be important to determine that the health states used in the measurement match those of your own study. These are some generic instruments used to measure health states, such as EuroQ_0L D5 or Rosser-kind-Welham Index, provide a utility rating that can be readily used to calculate QALYs.
3. Generally, the most accurate way to obtain utility values for your own study will be to measure them yourself. In this way one can tailor the valuation to the exact need of the study. This is the most resource intensive alternative, however.

There are three main techniques for assessing the preferences of respondents for health outcomes:

1. Rating Scale (RS)/ Visual Analogue Scale (VAS)
2. Time-Trade Off (TTO)
3. Standard Gamble (SG)

1. **Rating Scale Method:** RS methods (which are also called category rating/VAS) are commonly used by psychometricians. Appropriate applications of RS reflect contemporary developments in the cognitive sciences. Judgment/decision theory has been dominated by the belief that human decisions follow principles of optimality and rationality. Considerable research, however, has challenged the normative models that have attempted to demonstrate rational choice.

   Psychometric models divided the decision process into component parts. Health states are observed and categorized. Utilities are observed and categorized. Preferences are obtained as weights for these health states and the ratings apply to a particular point in time and are analogous to consumer choices under certainty.

   In the RS method, respondents are instructed to make a mark on the indicated scale (feeling thermometer) to describe where they feel the given health state ranks on a scale of death to perfect health.

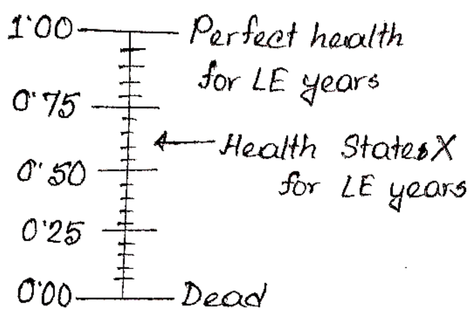


Figure-5: RS method

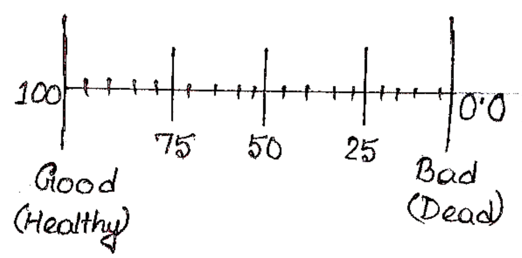


Figure-6: Visual Analogue Scale(VAS)

   The distance between the mark and these anchored states is to represent the degree of relative preference between the health states being valued. Several health states can be ranked on one rating scale. The distance between these marks is translated into a 0-1 scale and "utility" values are computed. Thus, a health state corresponding to a mark halfway between dead and perfect health would be assigned a utility value of 0.5.

   With VAS individuals are asked to indicate where on the line (Figure-6) between the Good and the Bad imaginable state they would rate/rank a certain health state. EuroQ$_0$L-5D is a form of VAS and so can also be used directly to value health states. Both RS/VAS measure preferences under conditions of certainty. Preference for chronic states or temporary health states can be measured on RS.

2. **Time Trade Off (TTO):**

The TTO method was developed specifically for use in health care by Tarrance, Thomas and Sackett (1972). The application of the TTO technique to a chronic state considered better than dead is shown in Figure-7. The subject is offered two alternatives:

i. state i for time t (LE of an individual with the chronic condition) followed by death;
ii. healthy for time X<t followed by death.

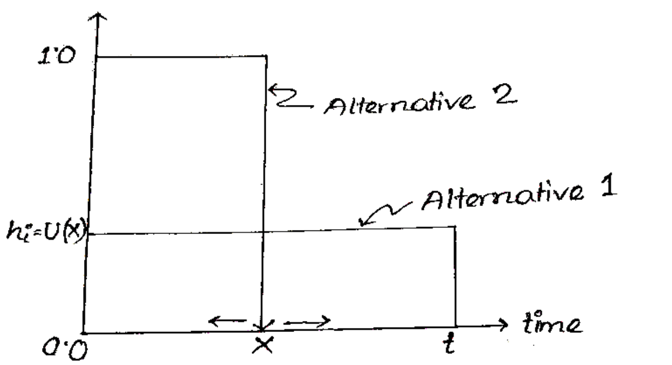


Figure 7: TTO for a chronic state preferred to death

It assesses preferences by asking a hypothetical question of willing to trade that health state (i) and its duration for the state of perfect health for a lesser number of years. The more debilitating the health state being valued the more years of life one would be willing to give up to attain perfect health for those years. If life expectancy (LE) is t years in health state i and someone will trade this for X<t years of life in perfect health, then the 'utility' of health state i is assumed to be represented by the fraction $\frac{X}{t}$. That is time X is varied until the respondent is indifferent between the two alternatives, at which point the required preference value for state i is given by

$$U(X)=h_i=\frac{X}{t}$$

Again, preferences for temporary health states can be measured relative to each other using the TTO as shown in Figure-8 below. As the RS, intermediate states i are measured relative to the best state (perfect health) and the worst state j (temporary state j). The respondent is given two alternatives:

i. temporary state i for time t (the time duration specified for the temporary state i) followed by perfect health,
ii. temporary state j for time X<t, followed by perfect health.

Note: VNM, Theory and Behavior under uncertainty attitudes toward risk.
The expected value of a lottery (P, W, W_L), where W_i are different wealth levels, is the sum of the outcomes, each multiplied by its prob of occurrence:

$$E[W]=PW_1+(1-P)*W_2 \quad \ldots\ldots\ldots\ldots\ldots(1)$$

A person is risk-neutral relative to a lottery if the utility of the expected value of the lottery equals the expected utility of the lottery, i.e., if

$$U\{E(W)\} = P*U(W_1) + (1-P)*U(W_2) \quad \ldots\ldots\ldots\ldots(2)$$

Risk neutrality implies a linear utility function: $U= \alpha+\beta W$, $\beta>0$

A person is risk averter relative to lottery if

$$U\{E(W)\} > P*U(W_1) + (1-P)*U(W_2) \quad \ldots\ldots\ldots\ldots(3)$$

Such a person prefers certain outcomes to uncertain ones with the same expected values. (3) implies that she has a quadratic utility function: $U=W-\alpha W^{\beta}$, $\alpha,\beta>0$

A person is risk lover relative to a lottery if

$$U\{E(W)\} < P*U(W_1) + (1-P)*U(W_2) \quad \ldots\ldots\ldots\ldots(4)$$

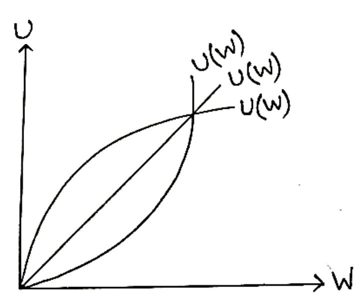
A risk lover will always take a fair bet. (4) implies that she has the following utility function: $U=\beta W^{-\alpha}$, $\alpha>0$

A measure of absolute risk aversion, r is provided by the negative ratio of the second derivation and first derivation of the utility function:

$$r = -\frac{u''(W)}{u'(W)} = -\frac{d\ lm\ u'(W)}{dW}$$

$$\text{when } r = \left\{ \begin{matrix} 0 \\ > \\ < \end{matrix} \right\} = \begin{matrix} \text{Risk neutrality} \\ \text{Risk aversion} \\ \text{Risk loving} \end{matrix}$$

The product (according to Arrow and Pratt (1964,1967), of rW provides a measure of relative risk aversion.



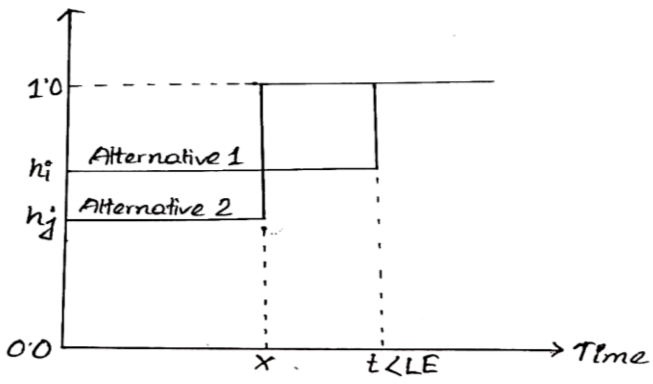


Figure 8: TTO for a temporary health state.

Time x is varied until the respondent is indifferent between the two alternatives such that:

$(1-h_i)*t = (1-h_j)*X$

$h_i*t - h_j*X = t-X$

$h_i= 1-(1-h_j)\frac{X}{t}$

If we set hj=0, then this reduces to hi=1-$\frac{X}{t}$. Figure 8 shows the basic format, but others variations are possible. State j may not be the worst state as long as it is any state worse than i.
It should be noted that TTO method measures preferences (as does RS or VAS) under conditions of certainty which may not be the true preferences or utilities elicited under conditions of uncertainty.

3. **Standard Gamble (SG):**
   The SG is the classical method of measuring cardinal preferences. It is based directly on the fundamental axioms (such as completeness, transitivity and continuity) of utility theory å lå von Neumann & O. Margenstern (1944). The method has been used extensively in the field of decision analysis.##

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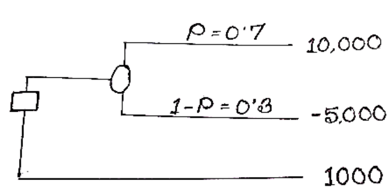
Let us assume that an individual is offered a lottery contract that there is 70-30 chance of winning 10,000 taka and losing 5000 taka. Now she is asked to trade the contract with someone else for 1000 taka. What is the utility of 1000 taka to her?
Assume that,

U(10000)= 1.0

U(-5000)=0.0

Then, U(1000)=0.7



The method can be used to measure preferences for chronic states but may vary somewhat depending upon whether or not the chronic state is preferred to death. For chronic states preferred to death the method is displayed in Figure-9.

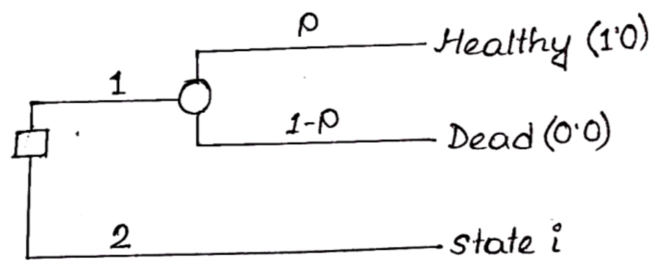


Figure 9: SG for a chronic health state preferred to death

The subject is offered two alternatives. Alternative 1 is a treatment with possible outcomes; either the patient is returned to normal health and lives for an additional t years with probability P, or the patient dies immediately with probability (1-P). Alternative 2 has the certain outcome of chronic state i for life (LE=t years). Now the prob P is varied until the respondent is indifferent between the two alternatives at which point the required preference value for state i is simply P; i.e., h_i=U(X)=P.

Preferences for temporary health states can be measured relative to each other using the SG method as shown in Figure 10.

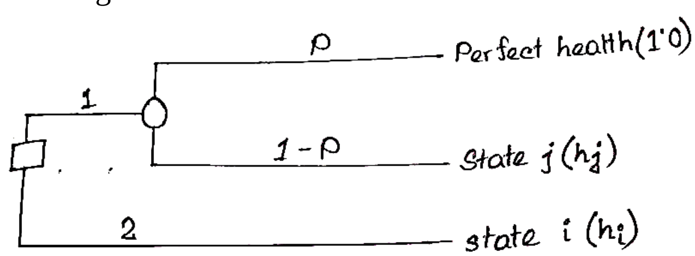


Figure 10: SG for a temporary health state.

In this case intermediate states i are measured relative to the best (perfect health) and the worst state (temporary state j). In this format the formula is

$$h_i=P+(1-P)h_j$$

where i is the state being measured and j is the worst state. If death is not considered in the use of utility, h_j can be set equal to zero and h_i values determined from the formula which then reduces to $h_i=P$.

However, if it is desired to relate these values to 0-1 scale, the worst of the temporary state (state j) must be redefined as a short duration chronic state (followed by death) and measured on the 0-1 scale by the technique described for the chronic states. This gives values for h_j which can then, in turn, be used in the above formula to find the value for the h_i.

Two problems are encountered in using SG method:

1. Treatment of most chronic diseases does not approximate the gamble. There is no product that will make arthritis better; nor is these one that is likely to cause immediate death (except the choice for euthanasia).
2. The cognitive demands of the task are high.

Psychological versus Economic models:

Health utility assessment has its root in the classical work of von Neumann and Morgenstern (1944). Then mathematical decision theory characterized how a rational individual should make decisions when faced with uncertain outcomes.
Tarrance and Feeny (1989) who reviewed the history of utility theory and its applications to health outcome assessment, argued that the use of the term 'utility theory' by VNM was unfortunate. Then reference to utility differs from the more common uses by economists that emphasize consumer satisfaction with commodities that are received with certainty. The 19th century philosophers and economists assumed the existence of cardinal (or interval level) utilities for these functions. A characteristic of cardinal utilities is that they can be averaged across individuals and ultimately used in aggregates as the basis of utilitarian social policy.

By the turn of the century, Pareto challenged the value of cardinal utilities and demonstrated that ordinal utilities could represent choice. Arrow (1951) further argued that there are inconsistencies in individual preferences under certainty and that meaningful cardinal preferences cannot be measured and may not even exist. As a result, many economists have come to doubt the value of preference ratings (Nord,1991).

Perhaps the most important statement against the aggregation of individual preferences was Arrow's impossibility theorem where he demonstrated that aggregate decisions can violate the apparent will of the individual decision makers.

Arrow's impossibility theorem may not be applicable to the aggregation of utilities in the assessment of QALYs for several reasons:

1. Utility expressions for QALYs are expressions of probability outcomes, not good received with certainty; VNM approach thus theoretically distinct from Arrows.
2. Arrow assumed that the metric underlying utility was not meaningful and not standardized across individuals. Substantial psychometric evidence now suggests that preferences can be measured using scales with meaningful interval or ratio properties. When cardinal(interval) utilities are used instead of ranking, many of the potential problems in the impossibility theorem are avoided (Keeny,1976).

Although different approaches to calculate QALYs are based on different underlying assumptions, they are nevertheless consistent with VNM notion of decision under uncertainty, in which probabilities and trade-offs are considered explicitly by the judge.

Economists have challenged the psychometric approaches, emphasizing that data obtained using RS/VAS cannot be aggregated. They acknowledged that RS may provide ordinal data but contend that they do not provide interval level information necessary for aggregations. These judgements under certainty are subject to all of the difficulties shown by Arrow (1951).
Armed with these arguments, economists have proposed SG or TTO methods as validity criteria for RS/VAS. The basic assumption is that methods that conform to VNM axioms assess true utility. SG serves as a 'gold standard' for evaluating other methods.

However, more exchange between economists and psychologists are needed to resolve the theoretical and practical difficulties of utility assessment.
A review of literature on CUA suggests that preferences obtained (by SG and TTO methods) can be explicitly considered in a CUA. A variety of studies have evaluated the generalizability, the validity, and the reliability of the preference measures.

The TTO method and Cardinal Utility:

The relationship between TTO method and cardinal utility theory is illustrated in Figure -11. The figure depicts the utility function with respect to LYs in FH and HS A. In the figure it is assumed that we are assessing the quality weight with TTO method for 10 years in HS A. To do this we find the number of years in FH that leads to the same utility as 10 years in HS A.

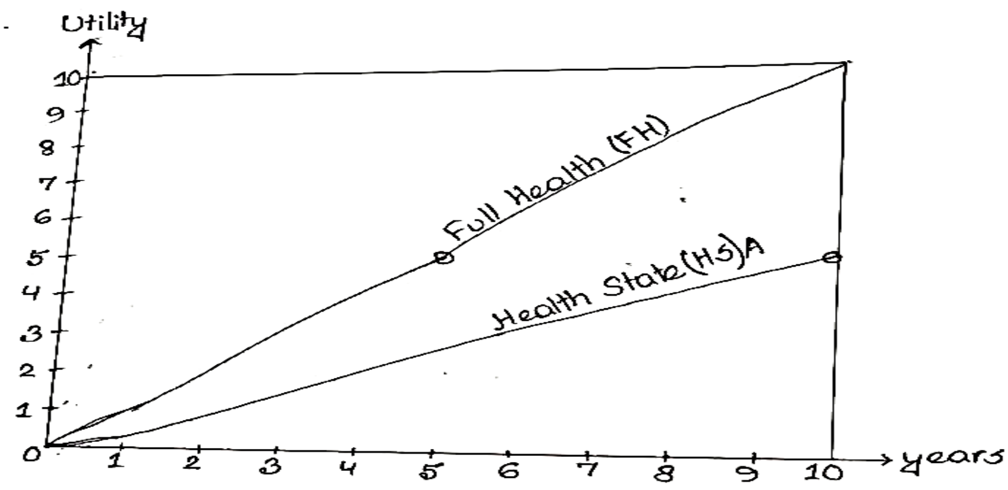


Figure 11: The TTO method and Cardinal Utility

In figure-11, 5 years in FH has the same utility as 10 years with HS A. The quality weight is derived by dividing the number of years in FH by the number of years in HS A, which gives a quality weight of 0.5 in the figure (5/10). This quality weight can be interpreted as showing that the utility is the same for 10 years in HS A and 0.5 of 10 years in FH, i.e., the quality weight is measured as the fraction of the number of healthy years.

The SG method and Cardinal utility:

The relationship between the SG method and cardinal utility theory is illustrated in Figure-12. The figure shows the utility function with respect to time in Full Health and the utility function with respect to time in health state A, which is a health state with less than full health (e.g., due to arthritis). In the figure it is assumed that we want to measure the quality weight for health state A using a time horizon of 10 years. With the SG method we then measure the utility of health state A for 10 years as a fraction of the utility of 10 years in FH.

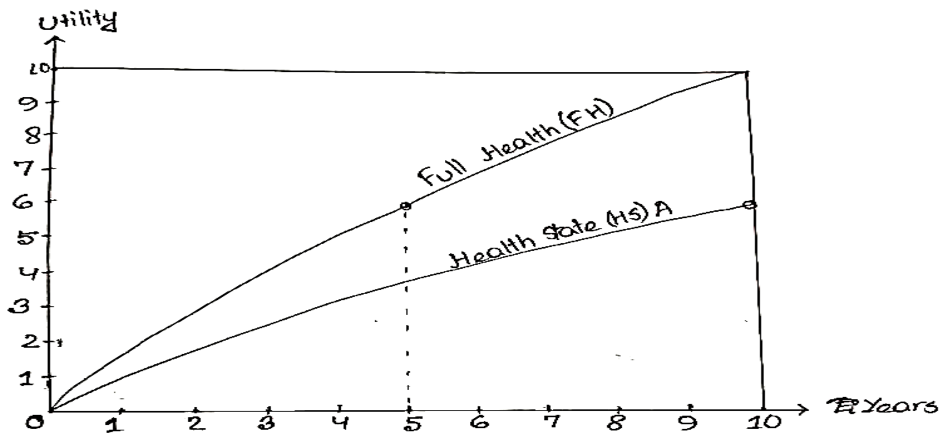


Figure-12: The SG method and Cardinal utility

The individual will be indifferent between 10 years in health state A and a gamble with 0.6 probability of FH for 10 years and 0.4 probability of immediate death. The expected utility of the gamble is (0.6*10+0.4*0) =6 and the expected utility of 10 years in health state A is 6(1.00*6). The SG method can thus be interpreted as measuring the fraction of the utility of FH for a health state, and this fraction is used as the quality weight (=0.6=$\frac{6}{10}$) to construct QALYs.

<u>Note</u>: In the figures 11 & 12, the quality weights differ between the TTO method and the SG method because the marginal utility of additional years is decreasing, due to the way the utility functions have been drowned. This means that 50% of the number of healthy years leads to more utility than 50% of the utility of healthy years (in the figure 50% of the number of years in FH leads to 60% of the utility of the same)